Analytical Chemistry Cumulative Exam

September 7, 2018

1. (5 points) Provide the analytical chemistry definition for *sensitivity*. Use illustrations as needed.

2. (10 points) What is the difference between *sensitivity* and *detection limit*? Explain how you would experimentally determine the detection limit for a new cytokine biosensor.

3. (5 points) What is a blank?

4. (5 points) Provide the concentrations with appropriate scientific notation in terms of M for the abbreviations pM and aM.

5. (15 points) Explain why the Bradford, Lowry, or biuret reactions would not be appropriate for quantitative measurements of cytokines in a blood sample.

6. (10 points) Explain the concept of fluorescence. Provide a Jablonski or energy diagram if it helps your argument.

7. (15 points) Define non-specific binding and explain why it is a problem with surface plasmon resonance methods of detection.

8. (35 points) Provide a brief summary of any additional knowledge you gained after reading this review article regarding spectroscopic techniques or the challenges of biomarker measurement that you did not know before.